60 single-best-answer questions | 90 minutes | 60 marks

Answer all questions. Choose one answer per question. Use a separate answer sheet.

1. An ECG marker is placed at the beginning of the P wave. Where must the second marker go to measure the PR interval?

A. At the beginning of the T wave.
B. At the beginning of the QRS complex.
C. At the end of the ventricular QRS complex.
D. At the end of the ventricular T wave.

2. Which response becomes particularly important for supporting pressure during sustained rather than momentary upright posture?

A. Renin-angiotensin effects and renal salt retention.
B. Suppression of aldosterone-driven sodium retention.
C. Persistent inhibition of all vascular reflexes.
D. Loss of sympathetic arteriolar responsiveness.

3. Why is the normal T wave generally broader than the QRS complex?

A. The T wave covers atrial and ventricular depolarisation together.
B. Ventricular depolarisation spreads more slowly than repolarisation.
C. The T wave corresponds to ventricular contraction rather than electricity.
D. Ventricular repolarisation usually takes longer than its depolarisation.

4. A person becomes angry during an argument, and both pulse and pressure rise. Which route can initiate this response?

A. Cortical signals acting through the hypothalamus on autonomic centres.
B. Atrial filling signals acting through the kidneys on increased urine output.
C. Cerebral CSF signals acting through the ventricles on the heart.
D. Platelet serotonin acting through capillaries on cardiac pacing.

5. Why can vision fail before consciousness during strong positive acceleration?

A. Retinal perfusion falls as blood shifts toward the feet.
B. The retina receives blood at higher pressure than the heart.
C. Intracranial pressure exceeds local arterial pressure.
D. Positive acceleration shifts blood toward the head.

6. A patient's right ventricular pumping fails and right atrial pressure rises. Which consequence is most likely?

A. A larger pressure gradient for systemic venous return.
B. Reduced congestion in peripheral systemic veins.
C. Reduced venous return gradient with congestion.
D. Complete disappearance of the jugular venous pulse.

7. Which combination correctly characterises isovolumetric ventricular contraction?

A. All valves closed, ventricular pressure rises, volume fixed.
B. AV valves open, semilunar valves open, volume rises.
C. AV valves open, semilunar valves shut, volume falls.
D. All valves closed, ventricular pressure falls, volume fixed.

8. A patient loses substantial albumin into urine during kidney disease. What Starling-force change is expected?

A. Increased red-cell viscosity causing filtration.
B. Reduced capillary permeability everywhere.
C. Increased plasma colloid osmotic pressure.
D. Reduced plasma oncotic pressure promotes oedema.

9. What produces the 'a' wave in a normal jugular venous pulse tracing?

A. Right atrial contraction.
B. Aortic valve opening.
C. Rapid ventricular ejection.
D. Pulmonary valve closure.

10. Why is perfusion of the left ventricular myocardium predominantly diastolic?

A. Ventricular contraction compresses intramural vessels.
B. The coronary ostia close during ventricular systole.
C. Aortic flow stops entirely during ventricular diastole.
D. The left ventricle receives no blood during relaxation.

11. Beta-1 adrenergic stimulation generally increases ventricular contractility through which sequence?

A. Blocked ryanodine receptors -> larger systolic calcium release.
B. Increased cAMP/PKA signalling -> greater calcium availability.
C. Increased vagal tone -> reduced pacemaker slope -> stronger contraction.
D. Reduced cAMP -> less calcium entry -> weaker contraction.

12. What principle allows an electromagnetic flowmeter to determine vascular flow?

A. Moving conductive blood induces voltage in a magnetic field.
B. An oxygen content difference equals flow independent of metabolism.
C. A cuff detects the onset of venous Korotkoff sounds.
D. A radioactive tracer measures only red-cell lifespan.

13. A person lying still stands up, then begins walking. What feature makes the calf-muscle pump effective?

A. Venule dilation without pressure gradients.
B. Capillary contraction during each footstep.
C. Deep vein compression with one-way valves.
D. Arterial compression with bidirectional valves.

14. Why may arterial blood pressure rise sharply during severe medullary ischaemia caused by raised intracranial pressure?

A. Direct vasomotor-centre inhibition greatly decreases total sympathetic outflow.
B. Increased vagal discharge is the main cause of the pressure rise.
C. Direct vasomotor-centre excitation greatly increases sympathetic outflow.
D. Increased atrial ANP release is the main cause of the pressure rise.

15. An atrial stretch receptor fires more during late atrial diastole when venous filling increases. Which receptor type fits best?

A. Type B, reflecting greater atrial filling.
B. An aortic body receptor, reflecting low oxygen.
C. A carotid sinus receptor, reflecting arterial stretch.
D. Type A, reflecting stronger atrial contraction.

16. A standard ECG uses ten skin electrodes but produces twelve electrical views. How is this possible?

A. Two extra electrodes inside the heart are automatically activated for recording.
B. Several leads are calculated from combinations of the same electrode voltages.
C. Each electrode produces one lead, with two more produced by blood-pressure sensors.
D. The limb electrodes are relocated to the chest during every cardiac cycle.

17. Why does positioning an accelerating pilot more horizontally improve tolerance to positive g?

A. It shortens the heart-to-brain hydrostatic column.
B. It increases the distance between the heart and brain.
C. It directs pooled blood toward the ankles.
D. It raises the capillary filtration rate in the legs.

18. Cardiac output falls by 20%, but total peripheral resistance rises by 25%. What happens to mean arterial pressure in the simplified MAP relation?

A. It remains approximately unchanged.
B. It decreases by approximately one quarter.
C. It increases by approximately one quarter.
D. It decreases by approximately one fifth.

19. A ventricular depolarisation vector runs almost perpendicular to the viewing axis of a lead. What should that lead record?

A. A small or biphasic ventricular deflection.
B. A tall downward ventricular deflection.
C. A tall upright ventricular deflection.
D. A prolonged ventricular depolarisation interval.

20. A vascular bed has a pressure difference of 80 mmHg and resistance of 20 mmHg?min/L. What is its flow?

A. 2 L/min.
B. 4 L/min.
C. 8 L/min.
D. 3 L/min.

21. Which time course of spontaneous recovery best fits the lecture after bleeding has been stopped?

A. Blood volume in weeks; haemoglobin in hours.
B. Both volume and haemoglobin fully recover in minutes.
C. Haemoglobin in 12-72 hours; volume in 4-5 weeks.
D. Volume in 12-72 hours; haemoglobin in 4-5 weeks.

22. In complete AV dissociation, the right atrium occasionally contracts against a closed tricuspid valve. What JVP abnormality may follow?

A. An isolated giant v wave after every beat.
B. Intermittent cannon a waves.
C. A steady nonpulsatile jugular column.
D. Loss of all visible venous pulsations.

23. During standing, head-level arterial pressure falls by 30 mmHg and cerebral venous pressure falls by 25 mmHg. How does arterial minus venous driving pressure change?

A. It falls by about 25 mmHg.
B. It rises by about 5 mmHg.
C. It falls by about 5 mmHg.
D. It rises by about 55 mmHg.

24. What effect does activating alpha-adrenoceptors have on vessels supplying the myocardium?

A. Coronary vasoconstriction.
B. Metabolic vasodilation.
C. Capillary oxygen extraction.
D. Greater cardiac muscle inotropy.

25. Redness spreads beyond a stroked patch of skin as the flare develops. What generates this spread?

A. Direct filtration through enlarged lymphatic vessels.
B. Venous thrombosis in the surrounding tissue.
C. Central spinal reflex contraction of the skin muscles.
D. Antidromic impulses in branches of a sensory axon.

26. A nurse notices ankle swelling after a long standing shift. Which immediate local change promotes this swelling?

A. Reduced capillary pressure decreases net filtration.
B. Lower interstitial pressure stops all capillary exchange.
C. Increased plasma oncotic pressure draws fluid outward.
D. Raised capillary pressure increases outward filtration.

27. During exercise, sympathetic activity increases throughout the body, yet working muscle receives much more blood. Which explanation is best?

A. A higher systemic pressure redirects flow equally among tissues.
B. Systemic arteriolar constriction raises every tissue's flow equally.
C. Local metabolic dilation outweighs constriction in active muscle.
D. Local metabolic constriction outweighs dilation in active muscle.

28. A low-frequency third heart sound occurs in early diastole. What ventricular event coincides with it?

A. Atrial depolarisation before systole.
B. Isovolumetric ventricular contraction.
C. Rapid passive ventricular filling.
D. Opening of the aortic valve.

29. A small additional deflection appears after the T wave in an otherwise interpretable trace. Which description fits the U wave?

A. Atrial depolarisation occurring before the next P wave.
B. A possible late ventricular repolarisation phenomenon.
C. Rapid ventricular depolarisation immediately before QRS.
D. A ventricular plateau interval between QRS and T.

30. Which substance passes relatively readily across the intact blood-brain barrier without a carrier?

A. Plasma albumin.
B. A protein-bound drug.
C. Carbon dioxide.
D. A large immunoglobulin.

31. What is the main purpose of wide changes in skin blood flow during heating or cooling?

A. To maintain constant oxygen uptake by skin cells.
B. To alter the heart's intrinsic pacemaker rate.
C. To adjust body heat gain and loss.
D. To equalise pulmonary and systemic arterial pressures.

32. Why does sustained systemic hypertension tend to raise myocardial oxygen demand?

A. Aortic pressure removes myocardial wall tension.
B. The left ventricle stops contracting during systole.
C. Coronary venous oxygen content necessarily becomes arterial.
D. The ventricle must generate higher pressure for ejection.

33. An echocardiogram shows poor ventricular contraction and a reduced ejection fraction. Which primary failure pattern does this support?

A. Stiff ventricular filling with preserved fraction.
B. Increased relaxation with low filling pressure.
C. Isolated systemic vasodilatation without pump disease.
D. Systolic dysfunction with impaired ejection.

34. Why is a distinct atrial repolarisation wave usually absent from a normal ECG?

A. Atrial repolarisation generates no electrical voltage differences.
B. The atria cannot repolarise until ventricular diastole begins.
C. Its small electrical signal is obscured by the ventricular QRS complex.
D. Atrial repolarisation occurs just before the SA node starts the next beat.

35. A measured QRS occupies 3.5 small squares at 25 mm/s. Which statement is supported by the lecture's usual duration range?

A. It lasts 0.10 s and is at the upper edge of the stated range.
B. It lasts 0.14 s and is longer than the stated usual range.
C. It lasts 0.07 s and is within the stated usual range.
D. It lasts 0.35 s and is longer than the stated usual range.

36. A Wiggers diagram shows the QRS complex before a steep ventricular pressure rise. What does the QRS represent?

A. Beginning of passive atrial filling.
B. Closure of the pulmonary venous openings.
C. Completion of ventricular repolarisation.
D. Electrical activation of the ventricles.

37. A patient becomes dizzy on standing after starting a sympatholytic drug. Which compensatory response is most directly weakened?

A. Contraction of calf muscles during walking.
B. Flow of blood through competent venous valves.
C. Dilatation of cerebral vessels by carbon dioxide.
D. Constriction of systemic arterioles and veins.

38. A patient develops rapidly increasing intracranial pressure after a brain haemorrhage. What first threatens cerebral perfusion?

A. Immediate loss of blood volume from the systemic arteries.
B. Compensatory dilation of brain vessels with increased driving pressure.
C. Increased cerebral perfusion despite rising intracranial pressure.
D. Compression of brain vessels and reduced driving pressure.

39. A vessel is 50 cm below heart level in an upright patient. Approximately how much hydrostatic pressure does gravity add?

A. About 19 mmHg.
B. About 39 mmHg.
C. About 77 mmHg.
D. About 100 mmHg.

40. What closes the aortic valve at the beginning of ventricular relaxation?

A. Venous pressure exceeds aortic pressure.
B. Atrial pressure rises above left ventricular pressure.
C. Aortic pressure exceeds ventricular pressure.
D. Left ventricular pressure rises above aortic pressure.

41. Which sequence best describes the classical triple response of Lewis?

A. White line, pallor, skin cooling.
B. Red line, flare and wheal.
C. Flare, complete blanching, no fluid leak.
D. Raised wheal, venous thrombosis, scar.

42. Which condition is most likely to depress myocardial contractility rather than enhance it?

A. Acute beta-1 adrenergic activation.
B. Raised circulating catecholamines.
C. Severe hypoxia with acidosis.
D. Increased calcium availability in cells.

43. A patient has had high arterial pressure for months. Why might baroreceptors fail to drive pressure back to the original baseline?

A. Their operating range resets around the sustained level.
B. Their stretch sensitivity increases indefinitely without adaptation.
C. Their afferent pathways turn into motor pathways over time.
D. Their signals stop entering the medulla during hypertension.

44. A very light stroke briefly leaves a pale line on the skin. How is this reaction best classified?

A. The wheal component of the triple response.
B. The red line of local capillary dilation.
C. The spread of an axon-reflex flare.
D. A brief local vasoconstrictor response.

45. Why can lying flat aggravate breathlessness in left ventricular failure?

A. It sharply lowers the volume returning from the legs.
B. Increased central blood volume raises pulmonary venous pressure.
C. It directly destroys red cells within the pulmonary capillaries.
D. Pulmonary circulation becomes independent of the left atrium.

46. How can fluid enter a blind-ended lymphatic capillary when interstitial pressure rises?

A. A continuous layer of smooth muscle forces plasma outward.
B. A central lymphatic heart pumps blood into tissues.
C. Rigid endothelial junctions prevent all protein entry.
D. Overlapping endothelial flaps admit interstitial fluid.

47. If the sinoatrial node fails, why may the heart still beat at a slower rate?

A. Cardiac nerves directly contract every ventricular cell.
B. The ventricles require no electrical signal to contract.
C. The aortic valve becomes an independent pacemaker.
D. Other conducting tissues retain latent automaticity.

48. A rhythm strip shows P waves at a steady rate and QRS complexes at another steady rate, with no consistent relationship. What is the best classification?

A. Sinus rhythm with a normal PR interval.
B. First-degree atrioventricular block.
C. Second-degree atrioventricular block.
D. Third-degree atrioventricular block.

49. A patient with raised intracranial pressure develops hypertension and bradycardia together. What best explains the slow pulse?

A. The hypertension directly activates cardiac beta-1 receptors.
B. The cardiac pacemaker stops responding to vagal signals.
C. The arterial baroreflex reacts to the sympathetic pressure rise.
D. The atrial volume reflex reacts to the sympathetic pressure rise.

50. Blood flow to the medullary vasomotor centre becomes severely reduced. Which local chemical change directly activates its sympathetic response?

A. Hyperoxia together with falling carbon dioxide.
B. Reduced hydrogen ions together with abundant oxygen.
C. Hypoxia together with accumulating carbon dioxide.
D. Increased oxygen delivery together with reduced acidity.

51. A patient has a large infarct affecting the lateral surface of one cerebral hemisphere. Which arterial territory is most likely?

A. Anterior cerebral artery.
B. Posterior cerebral artery.
C. Middle cerebral artery.
D. Anterior communicating artery.

52. Blood ejected from the right ventricle returns to the left atrium through which sequence?

A. Aorta -> coronary sinus -> pulmonary veins.
B. Aorta -> systemic capillaries -> venae cavae.
C. Pulmonary artery -> lung capillaries -> pulmonary veins.
D. Pulmonary veins -> systemic tissue capillaries -> venae cavae.

53. Arterial pressure increases suddenly and baroreceptor firing rises. What changes first in central reflex control?

A. NTS activation increases depressor and vagal signalling.
B. The SA node starts generating impulses without modulation.
C. NTS activation increases pressor and spinal sympathetic firing.
D. Sensory traffic to the medulla decreases despite added stretch.

54. A patient with severe tricuspid regurgitation has a prominent systolic jugular venous wave. Which wave becomes especially large?

A. a wave.
B. v wave.
C. x descent.
D. y descent.

55. A sudden increase in sympathetic venous tone can support cardiac output mainly through which immediate effect?

A. It closes arteriolar valves and stops peripheral outflow.
B. It dilates veins, increasing blood stored in the limbs.
C. It creates new circulating plasma within seconds.
D. Venoconstriction shifts stored blood centrally.

56. What most clearly differentiates myocardial infarction from brief stable angina?

A. Prolonged ischaemia can produce myocyte necrosis.
B. Infarction occurs only when heart rate is very slow.
C. Angina permanently destroys the affected myocardium.
D. Angina requires complete coronary artery rupture.

57. A patient's heart rate is 80 beats/min and stroke volume is 75 mL. What is cardiac output?

A. 4.8 L/min.
B. 6.0 L/min.
C. 7.5 L/min.
D. 8.0 L/min.

58. What helps maintain forward systemic arterial flow between heartbeats?

A. Venous valves directly generate aortic pressure.
B. The elastic recoil of large arteries sustains diastolic flow.
C. The diaphragm pumps blood through the aorta during inspiration.
D. Capillaries generate contractions that propel blood upstream.

59. A patient with coronary atheroma is comfortable at rest but develops exertional chest pain. Which explanation fits best?

A. The coronary artery becomes completely normal during exercise.
B. Restricted coronary flow reserve cannot meet increased demand.
C. Exercise lowers heart oxygen requirements below resting levels.
D. Coronary flow stops at rest but rest pain is absent.

60. At standard paper speed, a P wave spans 2.5 small squares. How does its duration compare with the stated normal range?

A. 0.04 s, shorter than the expected P-wave duration.
B. 0.10 s, within the expected P-wave duration.
C. 0.20 s, longer than the expected P-wave duration.
D. 0.25 s, longer than the expected P-wave duration.